

MD. ROFAZ HASAN RAFIU

AI Systems Engineer • Software Architect

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PROFESSIONAL SUMMARY

Computer Science & Engineering student at RUET specializing in **Production AI Systems, Local Small Language Models (SLMs), In-Browser Computer Vision**, and scalable backend infrastructure. Experienced in bridging machine learning inference with resilient distributed infrastructure: local SLM diagnostics (Ollama/Qwen), client-side WebAssembly OpenCV, deterministic latency fallbacks, and multi-tenant PostgreSQL backends.

EDUCATION

- Rajshahi University of Engineering & Technology (RUET) — B.Sc. in Computer Science & Engineering**2023 – 2027
 - Admitted through national combined engineering entrance examination (CUET, KUET, RUET) ranking in top merit percentile.
 - Coursework: Data Structures & Algorithms, Object-Oriented Design, Database Systems, Computer Networks, Discrete Math.
- St. Joseph Higher Secondary School — Higher Secondary Certificate (HSC), Science**2022
 - GPA: 5.00 / 5.00

TECHNICAL SKILLS

- AI & Machine Learning:** Local SLM Inference (Ollama, Qwen 2.5, Llama 3.2), Structured Outputs (JSON Schema/Zod), OpenCV.js (WebAssembly), TensorFlow.js (BlazeFace), OCR (Tesseract.js), Deterministic Fallbacks, Prompt Engineering.
- Backend & Systems:** Python (FastAPI, Flask), TypeScript, Go (1.22+), Node.js (Next.js 15 App Router), RESTful APIs, WebSockets (Socket.io), System Design, Microservices, OWASP API Security, JWT, Role-Based Access Control (RBAC).
- Databases & Storage:** PostgreSQL (Neon, Aiven), Prisma ORM 6.10, Redis (Caching, Rate-Limiting), SQLite, Dexie.js (IndexedDB PWA Storage).
- Cloud, DevOps & Tools:** Docker, NGINX, GitHub Actions (CI/CD), Edge Caching (SWR, CDN Cache-Control), Linux/Bash Scripting, Git, Jest.

FEATURED ENGINEERING PROJECTS

- Digital School & Exam Platform (Rofaz Academy)** [Next.js 15, TypeScript, PostgreSQL, Prisma, Ollama, TensorFlow.js]Live: rofazacademy.dev
 - Local SLM Diagnostic Pipeline:** Engineered an automated assessment diagnosis engine integrating local SLMs (Qwen 2.5 3B / Llama 3.2 via Ollama) with temperature 0.3 structured prompting to extract root-cause conceptual mistakes from raw submission data.
 - Deterministic Fallback Reliability:** Architected a sub-10ms algorithmic fallback calculating topic-level error distributions and model explanations directly from PostgreSQL, ensuring 100% diagnostic reporting availability during local inference saturation.
 - Edge Scalability & Caching:** Configured edge CDN caching headers (Stale-While-Revalidate) and SWR synchronization on Next.js 15 routes, reducing origin database connection load by up to 70% during peak exam traffic.
 - Multi-Tenant Architecture:** Designed relational PostgreSQL schemas with Prisma ORM 6.10, connection pooling, multi-role RBAC, and background PDF generation for admit cards, tabulation sheets, and question banks.
- FocusGuard — Distributed Cross-Platform Attention Platform** [Go 1.22, Docker, WebSockets, Chrome MV3, SQLite]github.com/rofazhasan/FocusGuard
 - Distributed Systems Daemon:** Built a central Go 1.22 service coordinating real-time WebSocket state streaming across desktop, mobile, and browser clients with synchronized aggregate screen-time budgets.
 - Zero-Latency Browser Compilation:** Compiled behavioral policy rules directly into native browser C++ filtering rules via Chrome Manifest V3 declarativeNetRequest, eliminating JavaScript evaluation latency.
 - Local-First Offline Resilience:** Engineered local SQLite policy caching with delta synchronization upon network reconnection; integrated Android VpnService UDP DNS sinkhole intercepting restricted domains with RFC 1035 NXDOMAIN.
- OMRView — In-Browser WebAssembly Computer Vision Engine** [React 19, TypeScript, Vite, OpenCV.js Wasm]github.com/rofazhasan/OMRView
 - WebAssembly CV Pipeline:** Built a client-side document processing engine running compiled OpenCV C++ inside browser WebAssembly threads, eliminating 100% of server-side image processing costs (\$0.00 infrastructure overhead).
 - Geometric Rectification:** Implemented Canny edge detection, Douglas-Peucker polygon approximation, and 4-point homography perspective warp (cv.warpPerspective) to rectify skewed smartphone camera captures in <600ms.
 - Adaptive Intensity Classifier:** Segmented bubble ROIs evaluating pixel intensity histograms against row baseline lighting to accurately grade answer sheets and detect multi-fill collisions.
- Atlas — macOS Native Financial Operating System** [Swift 6, SwiftUI, SwiftData, Vision OCR, CryptoKit, AppKit]github.com/rofazhasan/atlas
 - Cryptographic Vault:** Implemented zero-knowledge AES-256 GCM encryption with PBKDF2 key derivation and Touch ID / LocalAuthentication biometric gatekeeping.
 - Apple Vision OCR:** Built offline document scanner parsing tabular expense records directly into native SwiftData models.
- Krishi Bondhu AI — Visual Agricultural Diagnostic Assistant** [Flutter, Dart, Riverpod, sqflite, Hardware Camera]github.com/rofazhasan/Krishi_Bondhu_AI
 - Built an edge mobile camera acquisition pipeline standardizing leaf telemetry with local SQLite persistence and bilingual voice playback feedback for offline rural usability.
- Additional Open-Source Systems & Repositories**GitHub: github.com/rofazhasan
 - BioEntry Pro** (github.com/rofazhasan/BioEntry): Cross-platform biometric attendance system in Electron, React, and TensorFlow.js with WebGL/WASM acceleration.

- **Meal Manager V2** (github.com/rofazhasan/Meal-Management): Fullstack mess & hostel accounting ledger with automated expense allocation in React, TypeScript, and Prisma.
- **Study Companion** (github.com/rofazhasan/study-companion): AI-powered academic habit tracker in Flutter, Dart, Firebase, and Google Gemini AI.
- **sdms** (github.com/rofazhasan/sdms) & **FinTrack** (github.com/rofazhasan/FinTrack): Relational student management system and React Native personal wealth telemetry application.
- **Election Game** (github.com/rofazhasan/Election_Game): Political strategy simulation engine and complex state machine engineered in Godot Engine / GDScript.

ACHIEVEMENTS & LEADERSHIP

- **Creative Talent Hunt, Regional Champion:** Recognized by the Government of Bangladesh for outstanding academic, mathematical, and computer potential.
- **Technical Workshop Organizer:** Organized and taught hands-on peer workshops at RUET on Data Structures, Algorithms, and Object-Oriented System Architecture.
- **Competitive Programming & Hackathons:** Active participant in regional programming contests and national software hackathons.